STA302H1 Fall 2026 Final Project

Part 2: Poster Presentation

Overview

Due Date: Thursday, December 3, 11:59pm

- No Questions Asked Extension until: Sunday, December 6, 11:59pm (Please note the separate dropbox on Quercus). No additional extensions will be granted for any reasons, including corrupted file submissions. After the NQA deadline passes, late submissions will receive a penalty of 1% per hour, up to a maximum of 48 hours, after which a grade of 0 is assigned.
- Please note that if you intend to use the NQA extension time, you should not submit any documents prior to the original assigned deadline. If you submit to both dropboxes, the latest submission submitted before the assigned deadline will be graded.

Goal of the Assessment:

- Showcase your group's knowledge of the correct usage of methods and techniques from the course.
- Experience the process of conducting a complete linear regression analysis on real data.
- Think about the ethical responsibility of a statistician in the context of conducting a data analysis.
- Create a presentation-worthy research poster summarizing how data were analyzed to answer a proposed research question.

Learning Outcomes being Assessed:

- Conduct a complete analysis starting from a research question to model that reasonably answers said research question.
- Communicate the motivation, process, and results of an analysis in a poster presentation.
- Report the results of a full analysis of the assumptions and recommend a course of action.
- Defend the decision to apply a transformation, the choice of transformation, and whether the violated assumption(s) is(are) adequately corrected.
- Defend the choice of "best" model for a given dataset and research question.
- Critique hypothesis test/confidence interval conclusions from an analysis as it pertains to violated assumptions.

- Appropriately discuss the ethical considerations of opting in/out of using automated selection methods.
- Interpret the results of an analysis involving linear models for technical and non-technical audiences.

Summary of Instructions

Using only methods and techniques presented in this course throughout the term, you and your group will answer your proposed research question from Part 1 (or modified based on the feedback received) by building an appropriate linear regression model on your chosen dataset (you do not need to start with the preliminary model from Part 1). You will want to ensure that your model was selected/built using evidence from the literature, dataset, and statistical methods. You will also discuss the limitations of your analysis (e.g. ethical, data, methodological, etc.) and what could be done differently in an ideal situation.

Ultimately, your main goal is to answer your proposed research question from Part 1 using the most suitable linear regression model possible built on your chosen dataset. You can consider more predictors than in your preliminary model from Part 1 and/or you can reduce the number of predictors in your model based on statistical tests or other relevant information. You should ensure you have attempted to correct any violated assumptions and have assessed your assumptions at appropriate stages of your analysis. You should also perform full diagnostics on your chosen model, including checking for multicollinearity, problematic observations, and violated assumptions. Finally, you need to explicitly answer your proposed research question using the statistical tools learned in the course.

Your group will produce two deliverables:

1. A poster that summarizes or provides an overview of the key message and results of your analysis and could be understood as a standalone without supplemental materials. Your poster should focus on telling the overarching story/message of the analysis with more emphasis on plots and tables rather than text and on selection of important and relevant information to help readers see the message at a glance.
2. A recorded presentation where you present your poster, where you go into more detail, for example, about any technical aspects and methodologies.

We are assessing you on whether you can correctly apply the course content and document statistical decisions with corresponding evidence while effectively communicating these results in a concise but clear manner, as there is limited space on your poster. You've already had valuable practice in these areas through your coursework

so far. These skills are important for many academic and industry positions and will provide you with materials to use in a portfolio or list on a resume.

Opportunities to Showcase Your Work

We are also looking to nominate a few groups to enter into the Canadian ISLP Poster Competition to showcase the amazing work that UofT undergraduate students are doing in their courses. Winning posters receive a cash prize and these groups will go on to represent Canada in the international poster competition of the International Statistical Institute (ISI).

In Fall 2024, a group from STA302 won first prize for their final project poster.

Groups can opt into being considered for this competition when submitting your work, and the teaching team will select the best ones to be nominated. Your grade is NOT impacted by your decision to opt in or out of this competition, or by the results of the competition. We will not make any decisions about nominations until after the grading process is complete.

There are many other opportunities if you wish to showcase the work you are doing in this project, such as:

- You can turn your poster into a report and submit to the Undergraduate Statistics Class Project Competition.
- You can present your poster at the Canadian Statistics Student Conference coming this summer.

Poster Requirements

The poster is the overview of your whole analysis, from the question through what techniques you applied to the final answer to your question. Your focus should be on limiting the amount of text and selecting/creating figures/tables that contain the information needed to understand why this model answers your research question.

Content of Poster

The poster will contain the typical parts of a research poster presentation, including all of the following:

- **Motivation and Research Question:** Briefly state your research question and why this is an important problem or question to be studying (e.g., who will benefit most from research). Discuss why fitting a linear model to answer your research question is preferred over a simpler statistical technique. Can be supported by literature where applicable.

- **Data Collection:** Where did you source your data (with appropriate citation), how was it originally collected (e.g. sample size, reliability of source, are measurements accurate, method of collection, etc.)? How is data appropriate for answering research question (e.g. same population/ability to generalize to population of interest, contains meaningful information/variables, etc.).
- **Methods of Analysis:** Give the readers a rough idea of what methods you used to arrive at your final model, including whether you employed model selection techniques, model validation, diagnostics for violated assumptions or other specific issues and remedies to these, and what tools/methods were used for each (a simple flow diagram could be useful here to replace excessive text).
    - Within the methods section, please make sure you address the following question prompts: If you chose to use automated selection methods in your analysis, explain why you did not use manual selection methods. If you chose to use manual selection methods, explain why you did not use automated selection methods. Did you choose one for purely practical reasons?
- **Results and Conclusions:** Display final model giving particular emphasis to anything that helps answer your research question. Select figures and/or tables that highlight what prompted your final model to look as it does. Limit text to a discussion on how information in figures/tables led you to create final model and a concrete answer to your research question that is supported by the information presented.
- **Limitations:** a short discussion on at least one of: whether data were not entirely suitable and how, whether model is not entirely accurate or suitable for answering question (e.g. issues with diagnostics or assumptions), or whether model was not able to be validated (if appropriate). A suggestion is made for what could be done differently to improve the analysis overall.
- **References:** a mini-bibliography with citations for your data source and any other peer-reviewed work that was referenced in the poster. ONLY this section may be in a smaller font.

As part of the scaffolded nature of the final project, you and your group have already completed some of the components needed for this project. In your proposal, you motivated and proposed your research question, learned about your topic area through a literature summary, looked into the source of your dataset, summarized your dataset (numerically, graphically, in text), fit a preliminary model and assessed assumptions for violations. Consider the following:

- Use feedback received to revise the research question.

- Use your proposal to craft the introduction/motivation.
- Use your proposal to talk about data used, how it was collected, and its trustworthiness.
- Use your proposal as a guide on how to check and report violated assumptions.

Technical Requirements for the Poster

- Must **not** contain student names, group number, or university or class information.
- Include a title that reflects your research question and/or analysis.
- The maximum size is A1 (841 mm x 594 mm or 33.1 in x 23.4 in) and the maximum file size is 10 MB. For electronic posters, the font size should be such that if printed in A1, it would be readable from approximately 2 metres (7 feet) away. There is no minimum poster size.
- The data source must be cited in the poster (under references).
- Minimal text should be used (instead consider using figures/tables and including captions that highlight what you want your reader to know).
- Feel free to use colour and be creative in the layout and design of your poster.

Presentation Requirements

Each group will submit a single video recording presenting their poster. The presentation should give an overview of your entire research project, and any viewers of your presentation should understand what your processes were. This includes everything from what your research question is, the data you used, the methods and techniques you applied to answer your question, along with any discussion around your results. Rather than try to explain everything you have done during your presentation, try to focus on the most important aspects that tells your research story.

Please note the following requirements:

- The audience for your presentation will be other undergraduate science students, who may or may not have taken STA302H1. That is, you cannot expect the audience to understand the nuanced methods and details covered within STA302H1. However, you can expect the audience to have a general understanding of scientific methods and the research cycle, as well as a general understanding of quantitative methods. If it helps, you may pretend that you are presenting this poster at an undergraduate research fair.
    - The presentation portion of your project emphasizes your ability to communicate technical results to audience members who may not be as technically inclined as you. Your presentation should be able to convince the audience of the importance and impact of your work and results.

- The video can be in any format. For example, the following are acceptable formats:
    - A video of your group presenting a physical printed poster or your poster projected onto a screen, as if you are at an in-person poster presentation. All group members must be in view of the camera for the entire duration of your group's presentation.
    - A recording of your group presenting a digital poster over Zoom or Teams, as if you are at a virtual poster presentation. All group members must have cameras on for the entire duration of your group's virtual presentation.
    - A creative video that presents your poster as if you were posting a video on social media. For example, you may have a vertical-format (or horizontal-format) video and overlay yourselves speaking over different parts of your poster as you zoom in and out.
- The video of the poster presentation must be **no longer than 8 minutes long** at a **normal speaking pace**. That is, you **cannot** speed up your video or purposely speak quicker than typical to meet the length requirement. There is no minimum length requirement.
- Please ensure that your video is high quality, with clear audio and that we can clearly see which sections of the poster you refer to as you speak. It is acceptable if we cannot make out the text on the poster in your video (for example, if you are recording yourself present a physical poster) since we can reference the digital version of your poster when grading, but we need to know what section to refer to.
- Please upload your video to your UofT OneDrive, and ensure that it is viewable via a link. **Broken links, incorrect sharing permissions, or if the grader cannot access your video for any other reason will result in a grade of 0 for the presentation component.**
- All group members must participate and be visible in the video. Any group members who are not visible and/or do not participate in the video will receive a grade of 0.

What to Submit

Only ONE member of the group should submit ALL required components. A complete submission to Quercus will include:

File uploads:

- The poster, summarizing the key message and meeting the above technical requirements, saved as a PDF.
- The Rmd file containing all the code used to conduct the data analysis (this is not being explicitly graded, but we may need to reference it to understand your results).

Please include the following as three separate submission comments:

- A labelled link to your presentation video, uploaded to your UofT OneDrive, with the **shareable link** included as a **submission comment** on Quercus. Select the sharing permission "anyone in UofT with the link can access".
- A labelled link to the dataset used in the data analysis as CSV files, uploaded to your UofT OneDrive, with the **shareable link** included as a **submission comment** on Quercus. Select the sharing permission "anyone in UofT with the link can access".
- One of the below statements indicating whether your group would like to be considered for nomination to the Canadian ISLP Poster Competition
    - *"Yes, our group would like to be considered for entry into the competition, and if selected, we give permission for this work to be displayed at the Statistical Society of Canada (SSC) and International Statistical Institute (ISI) conferences and special events, in publications and promotional material, and in electronic format on the Internet."*
    - *"No, our group does not wish to be considered for entry into the competition."*

New to Creating Posters?

- The University of Toronto library has lots of information on How to Create a Poster, including a video tutorial for how to create it in Microsoft PowerPoint. You can find examples of posters submitted to the competition at the competition website, or examples of the different ways you can organize your poster.
    - The 9th floor of OPG has graduate student research posters for inspiration
    - An Overleaf template can be found here (you can change the size of the poster in the preamble (see the options for the beamerposter package), to fit the requirements for the project).
    - Some PowerPoint templates from Rice University are here.
- If you want to export any of your plots from RStudio and aren't sure how, see here.

Rubric

Criteria of Assessment	Grading Scale
Poster	
Research Question and Data: • The research question is explicitly stated and a clear motivation for this topic is provided. • The source of the data is explicitly stated/provided and why it is appropriate to use to answer the question is correctly stated. • A discussion on how the data were originally collected and why it may be trustworthy or reliable is provided.	Each criterion is evaluated on a 4-level scale: • **3.0 (Excellent):** All checklist requirements met with **high precision, depth, and strong statistical rigor**. Work at this level can be presented formally at a professional setting, within an organization broadly or with external partners. • **2.0 (Satisfactory):** All checklist requirements met, but the discussion or reasoning is **generic, surface-level, or lacks analytical depth**. Work at this level can be shared within your team or department at a professional setting. • **1.0 (Revision needed):** Meets **some** requirements, but has **noticeable omissions, errors, or superficial analysis**. • **0.0 (Incomplete):** Minimal attempt; **missing multiple key requirements** or shows improper application of concepts.
Analysis and Conclusions: • The methods/techniques employed to answer the research question are outlined. • An answer to the research question is provided which is consistent with the analysis and evidence presented, and course content. • Sufficient evidence is presented that allows the reader to understand how the research question was answered and to evaluate whether the process was correct. • Limitations of the analysis are correctly noted with a suggestion for improvement for future studies.	
Poster appearance: • The poster does not contain a burdensome amount of text that could have been replaced with the use of figures or tables. • Figures and tables chosen are clear and convey an intentional message • The poster is logical and understandable without the need for additional or supplemental materials.	
Organization • All required sections are present in the poster, and are properly labeled • References are written in an appropriate and consistent citation format. • All figures/tables are correctly labeled, have informative captions, meaningful titles and axis labels. Individual plots that are combined into a single figure are appropriately sized and easy to read.	

Criteria of Assessment	Grading Scale
Presentation	
Introduction • The research question is stated, along with motivation for the importance of this study. • A brief summary of what the peer-reviewed academic literature says about the problem is provided. • The inferential technique(s) from the course (e.g., estimated coefficients, confidence intervals, prediction intervals, hypothesis test, etc.) that will provide the answer to the research question is(are) mentioned and consistent with the research question statement.	Each criterion is evaluated on a 4-level scale: • **3.0 (Excellent):** All checklist requirements met with **high precision, depth, and strong statistical rigor**. Work at this level can be presented formally at a professional setting, within an organization broadly or with external partners. • **2.0 (Satisfactory):** All checklist requirements met, but the discussion or reasoning is **generic, surface-level, or lacks analytical depth**. Work at this level can be shared within your team or department at a professional setting. • **1.0 (Revision needed):** Meets **some** requirements, but has **noticeable omissions, errors, or superficial analysis**. • **0.0 (Incomplete):** Minimal attempt; **missing multiple key requirements** or shows improper application of concepts.
Analysis Methods and Key Decisions • All relevant assumptions are checked and well justified, and other model diagnostics have been performed. • Verification of assumptions and other model diagnostics are performed and described in an appropriate order. • Modifications to the model based on verifying assumptions and other model diagnostics are clearly outlined and well justified. • Modifications to the model based on hypothesis tests or other variable selection approaches taught in the course are described and are correct. • The choice of using manual vs automated model selection tools is clearly justified, and explains why one method is chosen over its alternatives, either on practical or ethical grounds.	
Conclusion and Limitations • An answer to the research question is provided which is consistent with the analysis described, evidence presented, and course content. • A brief discussion is provided on whether the answer to the research question is consistent or not with the literature from the introduction. • Limitations of the analysis (or lack of) are identified (e.g. extreme observations, violated assumptions,	

limited data, etc.) and the impact of these are identified and described in a real-world context.	
Pacing and Organization • Speakers spoke at a good pace (not purposely very quickly), and the video was not sped up or slowed down • The presentation was within the time limit • All group members are visible in the video, participated equally and were all knowledgeable about the topic, and could speak about the topic without needing to reference a script or notes. • The format of the presentation is appropriate for the research project, topic, and intended audience. The presentation is engaging and interesting.	
Presentation Messaging • The presentation was made at an appropriate level for the audience (undergraduate students) • The format of the presentation and video takes the intended audience into account, and the level of detail, delivery, and messaging matches the format of the video chosen. • The presentation was easy to follow and informative. The presentation clearly communicated results as well as their significance. • A clear message is communicated to the audience.	

Total Points: /27